DEEP LEARNING

Attention Deep Dive

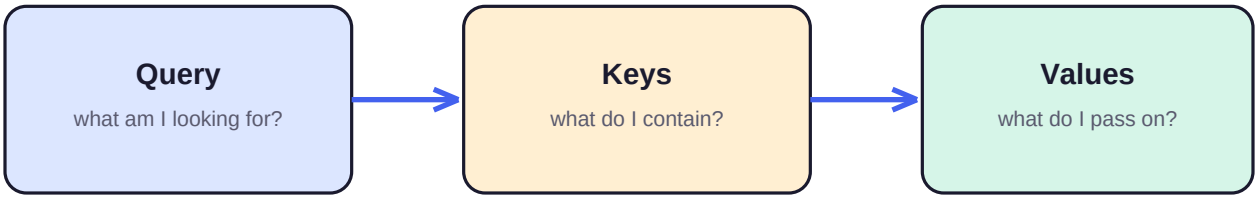
The mechanism behind transformers:
scaled dot-product attention, Q/K/V, masking, heatmaps.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Why Attention

Let every position look at every other position.



Attention = weighted lookup. The query scores every key; the scores weight the values.

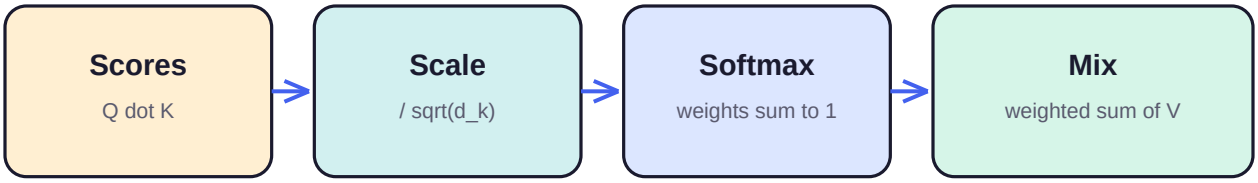
Unlike RNNs, attention sees the whole sequence at once — fully parallel, no forgetting curve.

STEP 2/5

Scaled Dot-Product Attention

Four operations, one formula.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(Q K^T / \sqrt{d_k} \right) V$$

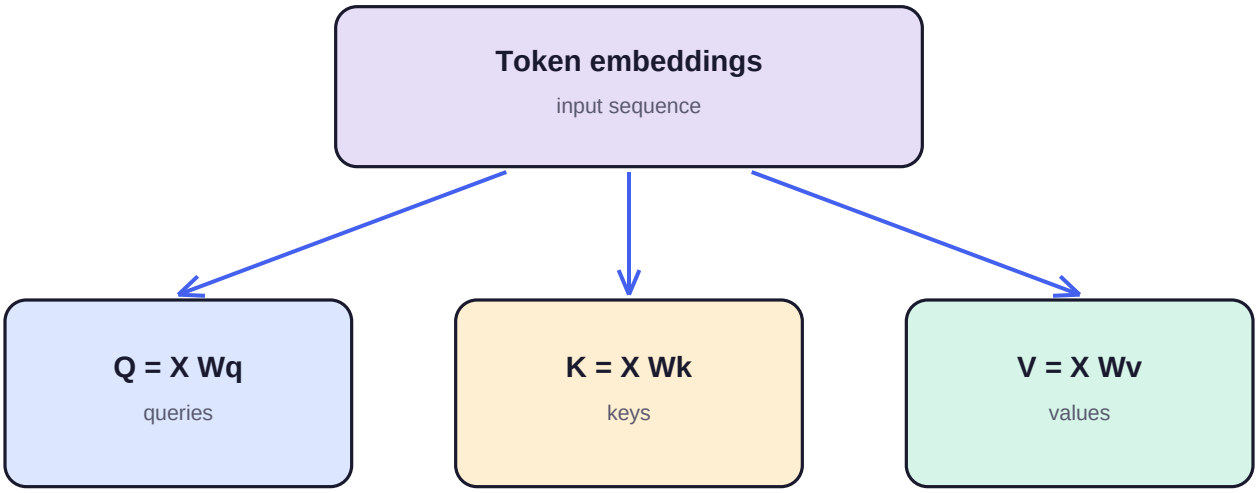


Scaling by sqrt(d_k) keeps dot products small so softmax stays in its sensitive range — without it, attention saturates.

STEP 3/5

Q, K, V Matrices

Three learned projections of the same embeddings.



Wq, Wk, Wv are learned. Multi-head attention runs several Q/K/V sets in parallel — each head learns a different relation.

STEP 4/5

Causal Masking

Decoders must not peek at the future.

Attention grid: rows = current position, columns = positions it may attend to.

Position 1: 'The'
attends to: The

Position 2: 'cat'
attends to: The, cat

'sat' blocked
future is masked

Before softmax, future scores are set to -infinity. After softmax they become exactly zero.

Masking is what makes next-token prediction honest: the model predicts each token from earlier positions only.

STEP 5/5

Reading Attention Heatmaps

Bright cells show strong attention weights.

Diagonal band
each word attends to neighbors — syntax lives here.
local attention:

Vertical stripe
many words attend to one token — often punctuation or [SEP]-like anchors.
global hub token:

- Rows sum to 1: a row spread thin means diffuse attention; one bright cell means focus.
- Different heads specialize: some track syntax, some track coreference, some track position.
- Caution: high attention weight does not prove causation — treat heatmaps as hints, not explanations.

STEP ✓I✓

Key Takeaways

Attention — Remember

- Attention = softmax(QK^T / sqrt(d_k)) V: queries score keys, weights mix values.
- Q, K, V are learned projections; multi-head runs several in parallel.
- Scaling prevents softmax saturation; masking enforces causality in decoders.
- Heatmaps reveal patterns (diagonal = local, stripes = hubs) but are not proof of reasoning.
- Parallel over the sequence is why transformers train so much faster than RNNs.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev